

Simple Exam Answers

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<https://chatgpt.com/c/6aa388a1-aef4-83ee-a0aa-51ff8522e55a>

Yes. Send me the question + its answer. I'll explain it in very simple exa...

26. PCA (Principal Component Analysis) Definition: PCA is a technique...

16. Data Cleaning Definition: Data cleaning is the process of removing ...

Yes. Send me **the question + its answer**.

I'll explain it in **very simple exam language**, using an easy way to remember it, so you can understand and reproduce it in the exam in one go.

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26. PCA (Principal Component Analysis)

Definition: PCA is a technique used to **reduce the number of variables/features** while keeping the important information.

Steps of PCA

1. Standardize the data.
2. Calculate the covariance matrix.
3. Find eigenvalues and eigenvectors.
4. Select the most important principal components.
5. Transform the data into the selected components.

Applications

- Image processing
- Data visualization
- Pattern recognition
- Feature reduction

Advantages

- Reduces data dimensions.
- Removes unnecessary information.
- Makes data easier to visualize and analyze.

Disadvantages

- Some information may be lost.
- Principal components are difficult to interpret.
- Results depend on proper data scaling.

Remember: Standardize → Covariance → Eigenvalues → Select → Transform

Histogram Graph

Definition: A histogram is a graph that shows the **frequency distribution of numerical data** using connected bars.

Characteristics

- Uses continuous numerical data.
- Data is divided into intervals called **bins**.
- Bars are usually connected.
- Height of each bar represents frequency.

Types

1. **Uniform Histogram** - values are almost equally distributed.
2. **Symmetric Histogram** - both sides have similar shape.
3. **Skewed Histogram** - data is concentrated more on one side.
4. **Bimodal Histogram** - has two main peaks.

Data Transformation

Why needed?

Data transformation is needed to **convert data into a suitable format** so that data mining algorithms can work effectively.

Different ways

1. **Smoothing** - removes noise from data.
2. **Aggregation** - combines data into a summary.
3. **Normalization** - brings values into a common range.
4. **Discretization** - converts continuous values into groups.
5. **Generalization** - replaces detailed data with higher-level data.

Remember: Smooth → Combine → Normalize → Group → Generalize

Drawbacks of Apriori Algorithm

1. Requires **many database scans**.

2. Generates a large number of candidate itemsets.
 3. Requires high **time and memory**.
 4. Becomes slow for large datasets.
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FP-Growth Algorithm

Definition: FP-Growth is an algorithm used to find **frequent itemsets without generating candidate itemsets**.

Steps

1. Scan the database and find frequent items.
2. Arrange items according to their frequency.
3. Build an **FP-Tree**.
4. Mine the FP-Tree to find frequent patterns.

Main advantage: It is faster than Apriori because it **does not generate candidate itemsets**.

Remember: Find → Arrange → Build Tree → Mine

43. Single Dimensional Association Rule

Definition: An association rule involving **only one dimension/attribute** is called a single-dimensional association rule.

Example:

Buys(Milk) → Buys(Bread)

It uses only the **“Buys”** dimension.

44. Multi Dimensional Association Rule

Definition: An association rule involving **two or more dimensions/attributes** is called a multi-dimensional association rule.

Example:

Age(20-30) AND Income(High) → Buys(Car)

Here, **Age, Income and Buys** are different dimensions.

45. Multilevel Association Rule

Definition: A multilevel association rule discovers relationships at **different levels of abstraction**.

Example:

- **Milk → Bread** - high level
- **Amul Milk → Brown Bread** - lower/more detailed level

Remember: Same relationship, different levels of detail.

46. Hybrid Association & Constraint-Based Association Rule

Hybrid Association

A rule that combines **different types of dimensions or attributes** is called hybrid association.

Example:

Age(20-30) AND Buys(Milk) → Buys(Bread)

Constraint-Based Association

It finds association rules by applying **user-defined constraints**.

Example:

Find only rules where **Support > 20%** and **Confidence > 70%**.

47. Classification as a Process

Definition: Classification is the process of **assigning data into predefined classes or categories**.

Process

1. Collect training data.
2. Build a classification model.
3. Test the model.
4. Use the model to classify new data.

Example:

Email → **Spam / Not Spam**

Remember: Train → Model → Test → Classify

48. Descriptive and Predictive Data Mining

Descriptive Data Mining

It explains **what has happened or what is present in the data**.

Example: Finding the average sales of a company.

Predictive Data Mining

It predicts **what may happen in the future** using existing data.

Example: Predicting next month's sales.

Remember:

Descriptive = What happened?

Predictive = What will happen?

49. Bayesian Classification / Naive Bayes

Definition: Bayesian classification is a classification method based on **Bayes' theorem**, which calculates the probability of a class based on given data.

```
genui{"learning_viz":{"type_id":"BAYES_THEOREM","locale_override":"en-IN"}}
```

Naive Bayes

Naive Bayes assumes that the features are **independent of each other**.

Example:

To classify an email as **Spam or Not Spam**, it checks words/features in the email and calculates which class has the higher probability.

Remember: Calculate probability → Compare classes → Choose highest probability

50. Decision Tree Induction

Definition: Decision Tree Induction is a method of building a **tree-like classification model** from training data.

Main parts

- **Root:** Starting attribute.
- **Branch:** Decision/result.
- **Leaf:** Final class.

Example:

Age → Income → **Buy / Not Buy**

Steps

1. Select the best attribute.
2. Split the data.
3. Repeat splitting.
4. Stop when the final classes are obtained.

51. Associative Classification

Definition: Associative classification is a classification method that uses **association rules to classify data**.

Types

1. **Class Association Rules (CARs)** - association rules where the consequent is a class label.
2. **Classification based on Association Rules** - uses multiple association rules to make classification decisions.

Example:

Age = Young AND Income = High → Class = Buys

52. Back Propagation

Definition: Back Propagation is a **supervised learning algorithm** used to train an artificial neural network by reducing the error between actual and predicted output.

Basic Steps

1. Give input to the network.
2. Calculate the output.
3. Find the error.
4. Send the error backward.
5. Update the weights.
6. Repeat until error becomes small.

Remember:

Input → Output → Error → Backward → Update → Repeat

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16. Data Cleaning

Definition: Data cleaning is the process of **removing or correcting incorrect, incomplete, and inconsistent data**.

Methods

1. **Missing value handling** – fill or remove missing values.
 2. **Noise removal** – remove unwanted/random errors.
 3. **Outlier detection** – identify unusual values.
 4. **Inconsistency correction** – correct conflicting data.
 5. **Duplicate removal** – remove repeated records.
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17. Process of Data Cleaning

Process

1. Identify errors in data.
2. Handle missing values.
3. Remove noisy data.
4. Detect and handle outliers.
5. Remove duplicates.
6. Check consistency and correct errors.

Tools

- **OpenRefine**
 - **Microsoft Excel**
 - **Python**
 - **R**
 - **SQL**
-

18. Methods

(a) Filling Missing Values

Missing values can be handled by:

1. **Ignore the record** – remove the record.
2. **Fill manually** – enter the correct value.
3. **Mean/Median/Mode** – use a suitable statistical value.
4. **Most probable value** – predict the missing value.

(b) Noisy Data Removal

Noise means **random errors or unwanted values** in data.

Methods:

1. **Binning** – replace values using bin mean/median/boundaries.
2. **Regression** – fit data to a regression line and identify errors.
3. **Clustering** – values far from clusters may be noise.

19. Outliers

Definition: Outliers are data values that are **very different from the other values**.

Example:

Marks = 45, 50, 52, 48, **99** → 99 may be an outlier.

Detection Methods

1. **Statistical method** – use mean, standard deviation or Z-score.
2. **Distance method** – values far from other data are outliers.
3. **Density method** – values in low-density areas are outliers.
4. **Clustering method** – values not belonging to any cluster may be outliers.

20. Data Integration

Definition: Data integration is the process of **combining data from multiple sources into one consistent dataset**.

Issues

1. **Schema differences** – different database structures.
 2. **Naming conflicts** – same attribute has different names.
 3. **Data type differences** – different formats/types.
 4. **Duplicate data** – same data appears more than once.
 5. **Value conflicts** – different values for the same object.
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21. Importance and Approaches of Data Integration

Importance

- Combines data from different sources.
- Provides a complete view of data.
- Removes duplicate and inconsistent data.
- Improves data analysis.

Approaches

1. **Data Warehousing** – store integrated data in one central warehouse.
2. **Data Federation** – access data from different sources through a common interface.
3. **ETL** – Extract, Transform and Load data into a common system.

Remember: ETL = Extract → Transform → Load

22. Correlation Analysis in Data Integration

Why important?

Correlation analysis helps identify **relationships between attributes** and helps detect redundant or duplicate information during integration.

Techniques

1. **Pearson Correlation** – measures linear relationship between numerical data.
 2. **Chi-Square Test** – checks relationship between categorical attributes.
 3. **Covariance** – shows how two variables change together.
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23. Entity Identification Problem

Definition: Entity identification means determining whether records from different sources **refer to the same real-world object**.

Example:

Database 1: **Rahul Kumar, 9876543210**

Database 2: **R. Kumar, 9876543210**

Both may represent the **same person**.

24. Tuple Duplication Problem

Definition: Tuple duplication occurs when **the same record appears multiple times** after combining datasets.

Example:

ID	Name
101	Rahul
101	Rahul

The same tuple is repeated, so it should be identified and removed.

25. Data Reduction

Definition: Data reduction is the process of **reducing the size of data while keeping important information**.

Techniques

1. **Data Cube Aggregation** – summarize data.
2. **Attribute Subset Selection** – remove unnecessary attributes.
3. **Dimensionality Reduction** – reduce number of dimensions, e.g. PCA.
4. **Numerosity Reduction** – replace data with a smaller representation.
5. **Data Compression** – represent data using less storage.

Remember: Aggregate → Select → Reduce Dimensions → Reduce Data → Compress

26. PCA

Definition: PCA reduces the **number of attributes/dimensions** while preserving important information.

Steps

1. Standardize data.
2. Calculate covariance matrix.
3. Find eigenvalues and eigenvectors.
4. Select important principal components.
5. Transform the data.

Applications

- Image processing
- Data visualization
- Pattern recognition
- Feature reduction

Advantages

- Reduces dimensions.
- Removes unnecessary information.
- Makes analysis easier.

Disadvantages

- Some information can be lost.
- Components can be difficult to interpret.
- Data usually needs proper scaling.

Remember: Standardize → Covariance → Eigenvalues → Select → Transform

27. Sampling

Definition: Sampling is the process of selecting a **smaller representative part of a dataset** for analysis.

Probability Sampling

Every item has a **known chance** of being selected.

1. **Simple Random Sampling** – every item has equal chance.
2. **Systematic Sampling** – select every kth item.
3. **Stratified Sampling** – divide data into groups and sample from each.
4. **Cluster Sampling** – divide data into clusters and select some clusters.

Non-Probability Sampling

Items are selected **without a known probability**.

1. **Convenience Sampling** – select easily available items.
 2. **Judgment Sampling** – researcher selects based on judgment.
 3. **Quota Sampling** – select a fixed number from each group.
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28. Attribute Subset Selection

Definition: Attribute subset selection means selecting **only the useful attributes** and removing unnecessary ones.

(a) Stepwise Forward Selection

1. Start with **no attributes**.
2. Add the best attribute.
3. Continue adding useful attributes.
4. Stop when no improvement occurs.

Remember: Empty → Add → Add → Stop

(b) Stepwise Backward Elimination

1. Start with **all attributes**.
2. Remove the least useful attribute.
3. Continue removing unnecessary attributes.
4. Stop when further removal reduces performance.

Remember: All → Remove → Remove → Stop

(c) Combination

1. Start with some/all attributes.
2. Add useful attributes.
3. Remove unnecessary attributes.
4. Repeat until the best subset is obtained.

Remember: Add + Remove

29. Histogram Graph

Definition: A histogram shows the **frequency distribution of numerical data** using connected bars.

Characteristics

- Used for numerical data.
- Data is divided into intervals called **bins**.
- Bars are connected.
- Bar height represents frequency.

Types

1. **Uniform** – values are almost equally distributed.
2. **Symmetric** – both sides have similar shape.
3. **Skewed** – data is concentrated toward one side.

4. **Bimodal** – has two main peaks.
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30. Data Transformation

Definition: Data transformation converts data into a **suitable format for data mining and analysis**.

Methods

1. **Smoothing** – removes noise.
2. **Aggregation** – combines data.
3. **Normalization** – brings values to a common scale.
4. **Discretization** – converts continuous values into groups.
5. **Generalization** – converts detailed data into higher-level data.

Remember: Smooth → Aggregate → Normalize → Discretize → Generalize

31. Data Normalisation

Definition: Normalization changes data values to a **common scale**.

Given: **200, 300, 400, 600, 1000**

(a) Min-Max Normalization for 400

Formula:

$$\text{New value} = (x - \min) / (\max - \min)$$

$$= (400 - 200) / (1000 - 200)$$

$$= 200 / 800$$

$$= \mathbf{0.25}$$

(b) Z-Score Normalization for 300

Formula:

$$\mathbf{Z} = (x - \text{mean}) / \text{standard deviation}$$

$$\text{Mean} = \mathbf{500}$$

$$\text{Standard deviation} \approx \mathbf{282.84}$$

$$Z = (300 - 500) / 282.84 \\ = -0.707$$

(c) Decimal Scaling for 600

Formula:

$$\text{New value} = x / 10^k$$

Maximum value = 1000, so **k = 4** to make all values less than 1.

$$600 / 10000 = 0.06$$

32. Techniques of Data Normalisation

1. **Min-Max Normalization** – converts values to a specified range such as 0 to 1.
2. **Z-Score Normalization** – uses mean and standard deviation.
3. **Decimal Scaling** – moves the decimal point by dividing by powers of 10.

Remember: Min-Max → Z-Score → Decimal

33. Concept Hierarchy Generation

Definition: Concept hierarchy organizes data from **low-level detailed values to high-level general values**.

Example:

Ahmedabad → Gujarat → India → Asia

It helps in **data generalization and analysis at different levels**.

34. Data Aggregation

Definition: Data aggregation is the process of **combining detailed data into summarized data**.

Example:

Daily sales → **Monthly sales → Yearly sales**

Working

1. Collect detailed data.

2. Group related data.
3. Apply an aggregation function.
4. Produce summarized data.

Types

- **Sum**
 - **Average**
 - **Count**
 - **Minimum**
 - **Maximum**
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35. Data Discretization

Definition: Data discretization converts **continuous numerical values into a small number of intervals or categories.**

Example:

Age values:

10, 15, 25, 35, 50

Can become:

- **Young:** 0-20
- **Adult:** 21-40
- **Old:** 41+

Common methods

1. **Binning**
2. **Histogram analysis**
3. **Clustering**
4. **Decision-tree-based discretization**

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